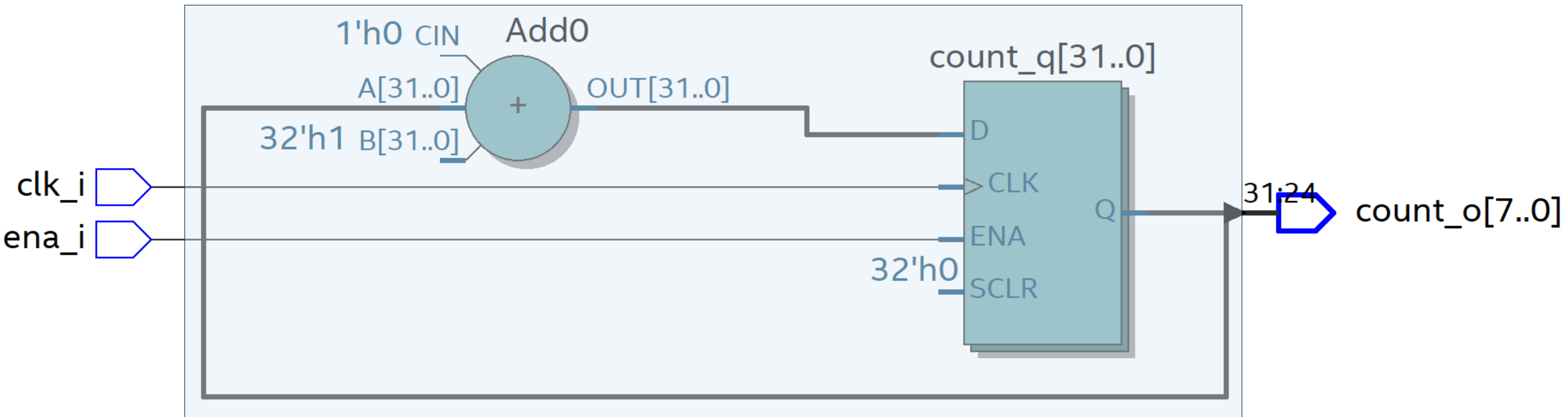
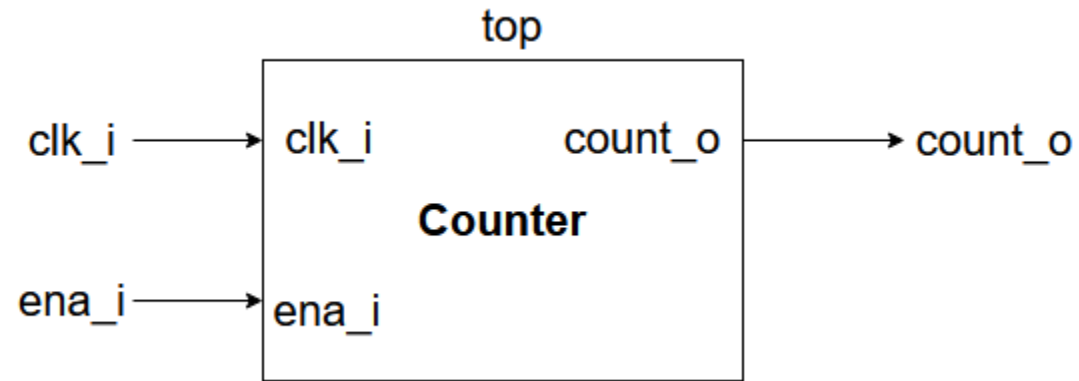


Counter example

Timing & Power

Analysis

Counter RTL



Counter – Area summary

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Flow Summary

<<Filter>>

Flow Status	Successful - Thu Apr 30 16:15:30 2026
Quartus Prime Version	21.1.0 Build 842 10/21/2021 SJ Lite Edition
Revision Name	Test
Top-level Entity Name	counter
Family	Cyclone IV E
Device	EP4CE115F29C7
Timing Models	Final
Total logic elements	32 / 114,480 (< 1 %)
Total registers	32
Total pins	10 / 529 (2 %)
Total virtual pins	0
Total memory bits	0 / 3,981,312 (0 %)
Embedded Multiplier 9-bit elements	0 / 532 (0 %)
Total PLLs	0 / 4 (0 %)

Counter - Setup time

Slow 1200mV 85C Model

Command Info		Summary of Paths						
	Slack	From Node	To Node	Launch Clock	Latch Clock	Relationship	Clock Skew	Data Delay
1	16.269	count_q[1]	count_q[30]	clk	clk	20.000	-0.080	3.669

Path #1: Setup slack is 16.269

Path Summary	Statistics	Data Path	Waveform	Extra Fitter Information
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Data Arrival Path

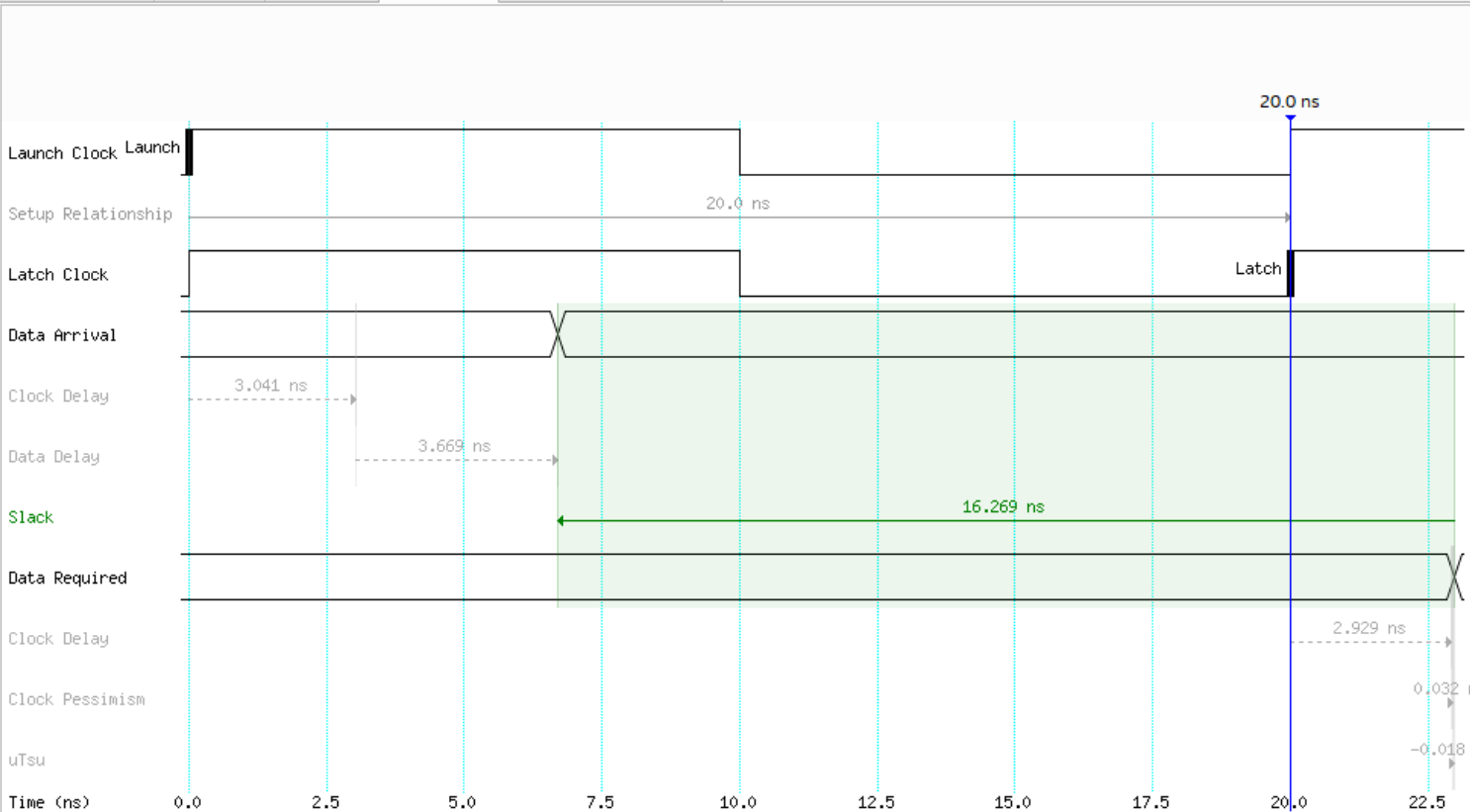
	Total	Incr	RF	Type	Fanout	Location	Element
1	0.000	0.000					launch edge time
2	3.041	3.041					clock path
1	0.000	0.000					source latency
2	0.000	0.000			1	PIN_Y2	clk_i
3	0.000	0.000	RR	IC	1	IOIBUF_X0_Y36_N15	clk_i~input i
4	0.720	0.720	RR	CELL	1	IOIBUF_X0_Y36_N15	clk_i~input o
5	0.907	0.187	RR	IC	1	CLKCTRL_G4	clk_i~inputclkctrl inclk[0]
6	0.907	0.000	RR	CELL	32	CLKCTRL_G4	clk_i~inputclkctrl outclk
7	2.443	1.536	RR	IC	1	FF_X88_Y72_N3	count_q[1] clk
8	3.041	0.598	RR	CELL	1	FF_X88_Y72_N3	count_q[1]
3	6.710	3.669					data path

Data Required Path

	Total	Incr	RF	Type	Fanout	Location	Element
1	20.000	20.000					latch edge time
2	22.961	2.961					clock path
1	20.000	0.000					source latency
2	20.000	0.000			1	PIN_Y2	clk_i
3	20.000	0.000	RR	IC	1	IOIBUF_X0_Y36_N15	clk_i~input i
4	20.720	0.720	RR	CELL	1	IOIBUF_X0_Y36_N15	clk_i~input o
5	20.900	0.180	RR	IC	1	CLKCTRL_G4	clk_i~inputclkctrl inclk[0]
6	20.900	0.000	RR	CELL	32	CLKCTRL_G4	clk_i~inputclkctrl outclk
7	22.374	1.474	RR	IC	1	FF_X88_Y71_N29	count_q[30] clk
8	22.929	0.555	RR	CELL	1	FF_X88_Y71_N29	count_q[30]
9	22.961	0.032					clock pessimism removed

Path #1: Setup slack is 16.269

Path Summary	Statistics	Data Path	Waveform	Extra Fitter Information
--------------	------------	-----------	----------	--------------------------



Counter - Setup time paths

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 - Setup Transfers**
 - Hold Transfers
- Report TCCS
- Report RSKM
- Unconstrained Paths

Setup Transfers

<<Filter>>

	From Clock	To Clock	RR Paths	FR Paths	RF Paths	FF Paths
1	clk	clk	528	0	0	0

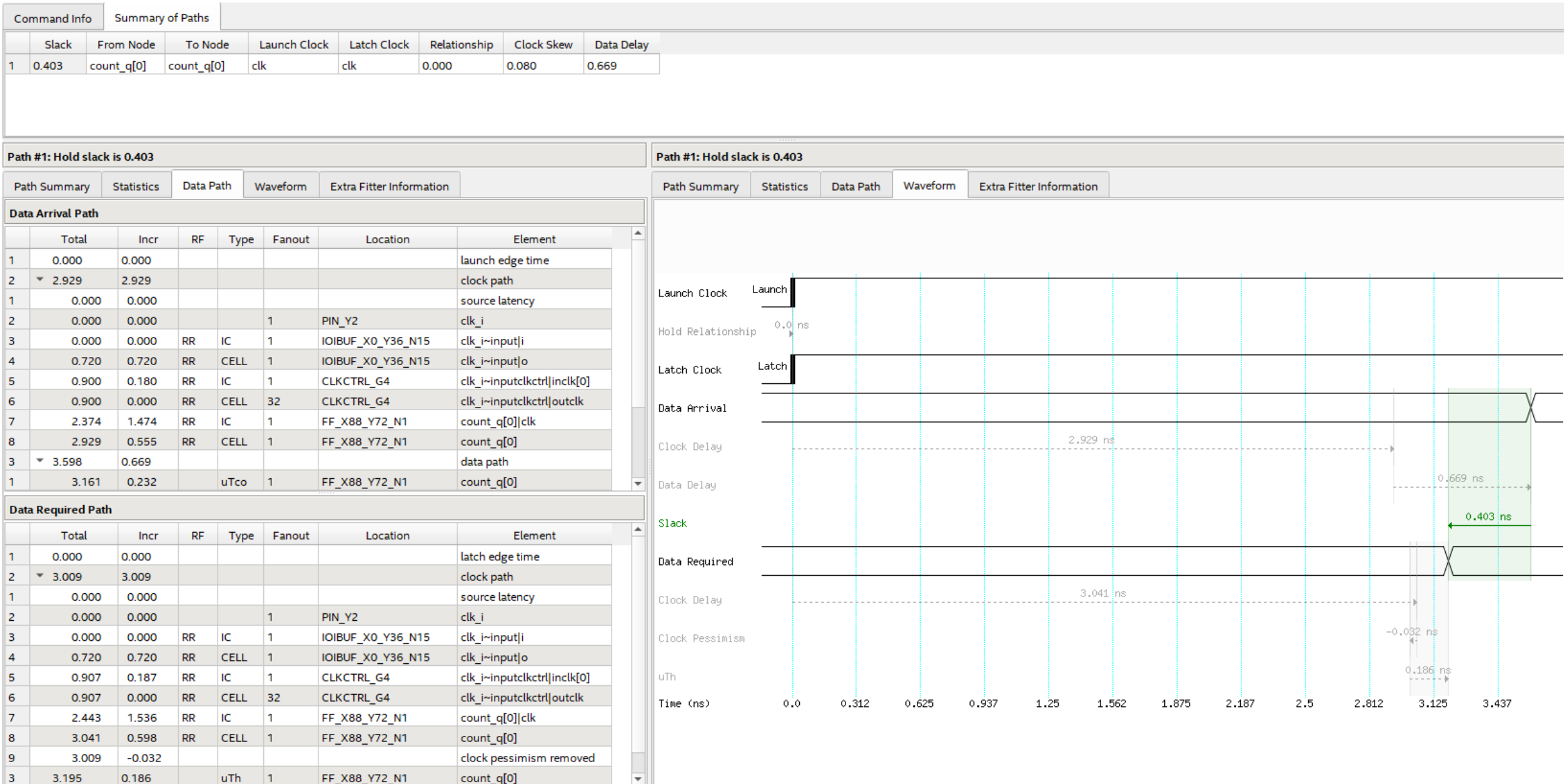
RR Paths stands for **Rising-edge-to-Rising-edge Paths**. It is a classification to categorize RR data paths based on the timing edges of the source (launch) and destination (capture) registers.

Explanation: in the counter example, the circuit's synthesis gives 32 series of source register to destination register routes {1, 2, ..., 32} for the Adder's c_{in} to c_{out} lines. The sum of this arithmetic series is $RR\ Paths = \frac{n(a_1 + a_n)}{2} = \frac{32(1 + 32)}{2} = 528$

Entries labeled "false path" only account for clock-to-clock false paths and not path-based false paths. As a result, actual path counts may be lower than reported.

Falling-edge-to-Rising-edge Paths

Counter - hold time



Counter - hold time paths

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 - Setup Transfers
 - Hold Transfers**
- Report TCCS
- Report RSKM
- Unconstrained Paths

Hold Transfers

<<Filter>>

	From Clock	To Clock	RR Paths	FR Paths	RF Paths	FF Paths
1	clk	clk	528	0	0	0

Entries labeled "false path" only account for clock-to-clock false paths and not path-based false paths. As a result, actual path counts may be lower than reported.

Counter - clocks

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 - Minimum Pulse Width Summary

Clocks

<<Filter>>

	Clock Name	Type	Period	Frequency	Rise	Fall	Duty Cycle	Divide by	Multiply by	Phase	Offset	Edge List	Edge Shift	Inverted	Master	Source	Targets
1	clk	Base	20.000	50.0 MHz	0.000	10.000											{ clk_i }

Counter – fmax summary

Compilation Report - Test X

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Slow 1200mV 85C Model Fmax Summary

<<Filter>>

	Fmax	Restricted Fmax	Clock Name	Note
1	268.02 MHz	250.0 MHz	clk	limit due to minimum period restriction (max I/O toggle rate)

This panel reports FMAX for every clock in the design, regardless of the user-specified clock periods. FMAX is only computed for paths where the source and destination registers or ports are driven by the same clock. Paths of different clocks, including generated clocks, are ignored. For paths between a clock and its inversion, FMAX is computed as if the rising and falling edges are scaled along with FMAX such that the duty cycle (in terms of a percentage) is maintained. Altera recommends that you always use clock constraints and other slack reports for sign-off analysis.

Counter - Power summary

Compilation Report - Test X

Power Analyzer Tool X

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- Operating Conditions Used
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- Core Dynamic Thermal Power Dissipati
- Current Drawn from Voltage Supplies
- Confidence Metric Details
- Signal Activities
- Messages

Power Analyzer Summary

<<Filter>>

Power Analyzer Status	Successful - Thu Apr 30 17:25:13 2026
Quartus Prime Version	21.1.0 Build 842 10/21/2021 SJ Lite Edition
Revision Name	Test
Top-level Entity Name	counter
Family	Cyclone IV E
Device	EP4CE115F29C7
Power Models	Final
Total Thermal Power Dissipation	136.22 mW
Core Dynamic Thermal Power Dissipation	1.71 mW
Core Static Thermal Power Dissipation	98.48 mW
I/O Thermal Power Dissipation	36.03 mW
Power Estimation Confidence	Low: user provided insufficient toggle rate data

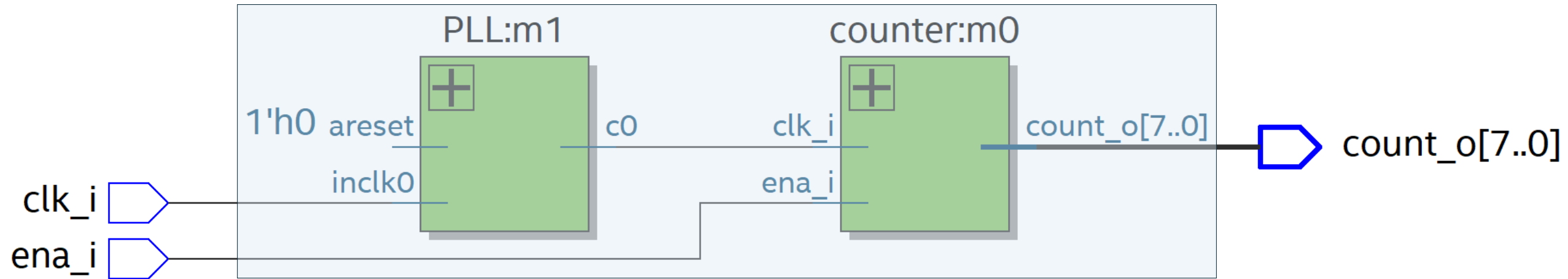
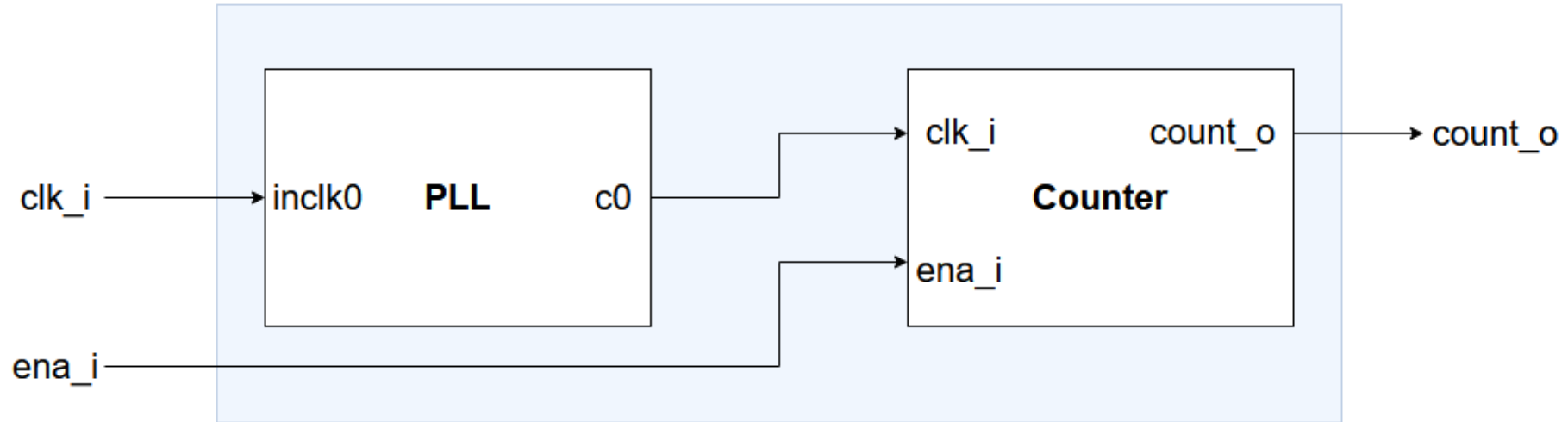
PLL-Counter example

Timing & Power

Analysis

Counter_pll RTL

top



Counter_pll - Area Summary

Compilation Report - Test X

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Flow Summary

<<Filter>>

Flow Status	Successful - Thu Apr 30 17:44:08 2026
Quartus Prime Version	21.1.0 Build 842 10/21/2021 SJ Lite Edition
Revision Name	Test
Top-level Entity Name	top
Family	Cyclone IV E
Device	EP4CE115F29C7
Timing Models	Final
Total logic elements	32 / 114,480 (< 1 %)
Total registers	32
Total pins	10 / 529 (2 %)
Total virtual pins	0
Total memory bits	0 / 3,981,312 (0 %)
Embedded Multiplier 9-bit elements	0 / 532 (0 %)
Total PLLs	1 / 4 (25 %)

Counter_pll - clocks

Compilation Report - Test X

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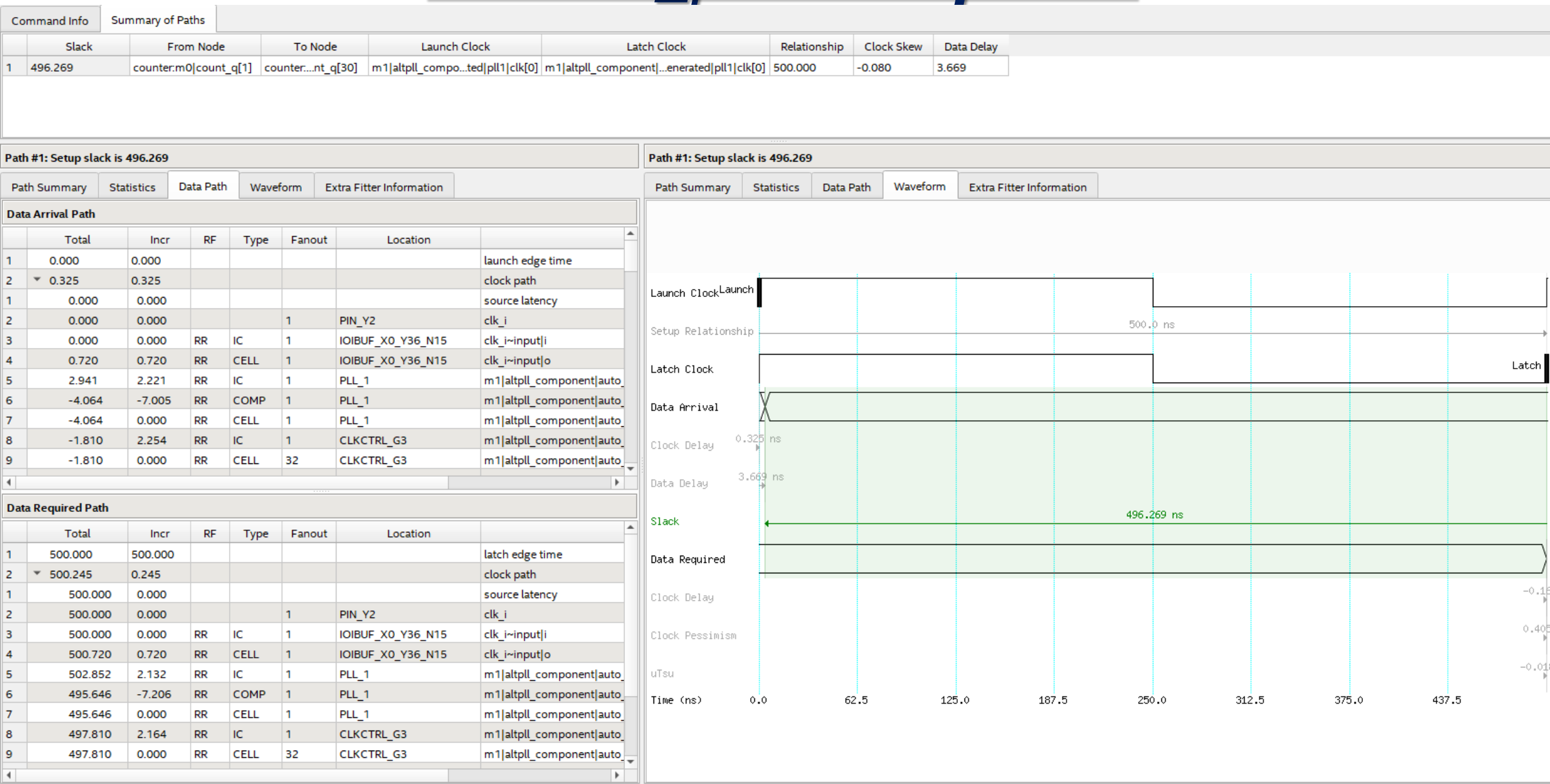
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Clocks

<<Filter>>

	Clock Name	Type	Period	Frequency	Rise	Fall	Duty Cycle	Divide by	Multiply by	Phase	Offset	Edge List	Edge Shift	Inverted	Master	Source	Targets
1	clk	Base	20.000	50.0 MHz	0.000	10.000											{ clk_i }
2	m1 altpll_com...d pll1 clk[0]	Generated	500.000	2.0 MHz	0.000	250.000	50.00	25	1					false	clk	m1 ...[0]	{ m1...[0] }

Counter_pll - Setup time



Counter_pll - hold time

Slow 1200mV 85C Model

Command Info

Summary of Paths

	Slack	From Node	To Node	Launch Clock	Latch Clock	Relationship	Clock Skew	Data Delay
1	0.403	counter:m0 count_q[0]	counter:m0 count_q[0]	m1 altpll_componen...erated pll1 clk[0]	m1 altpll_compon...ated pll1 clk[0]	0.000	0.080	0.669

Path #1: Hold slack is 0.403

Path Summary

Statistics

Data Path

Waveform

Extra Fitter Information

Data Arrival Path

	Total	Incr	RF	Type	Fanout	Location
1	0.000	0.000				launch edge time
2	-0.160	-0.160				clock path
1	0.000	0.000				source latency
2	0.000	0.000			1	PIN_Y2
3	0.000	0.000	RR	IC	1	IOIBUF_X0_Y36_N15
4	0.720	0.720	RR	CELL	1	IOIBUF_X0_Y36_N15
5	2.852	2.132	RR	IC	1	PLL_1
6	-4.354	-7.206	RR	COMP	1	PLL_1
7	-4.354	0.000	RR	CELL	1	PLL_1
8	-2.190	2.164	RR	IC	1	CLKCTRL_G3
9	-2.190	0.000	RR	CELL	32	CLKCTRL_G3

Data Required Path

	Total	Incr	RF	Type	Fanout	Location
1	0.000	0.000				latch edge time
2	-0.080	-0.080				clock path
1	0.000	0.000				source latency
2	0.000	0.000			1	PIN_Y2
3	0.000	0.000	RR	IC	1	IOIBUF_X0_Y36_N15
4	0.720	0.720	RR	CELL	1	IOIBUF_X0_Y36_N15
5	2.941	2.221	RR	IC	1	PLL_1
6	-4.064	-7.005	RR	COMP	1	PLL_1
7	-4.064	0.000	RR	CELL	1	PLL_1
8	-1.810	2.254	RR	IC	1	CLKCTRL_G3
9	-1.810	0.000	RR	CELL	32	CLKCTRL_G3

Path #1: Hold slack is 0.403

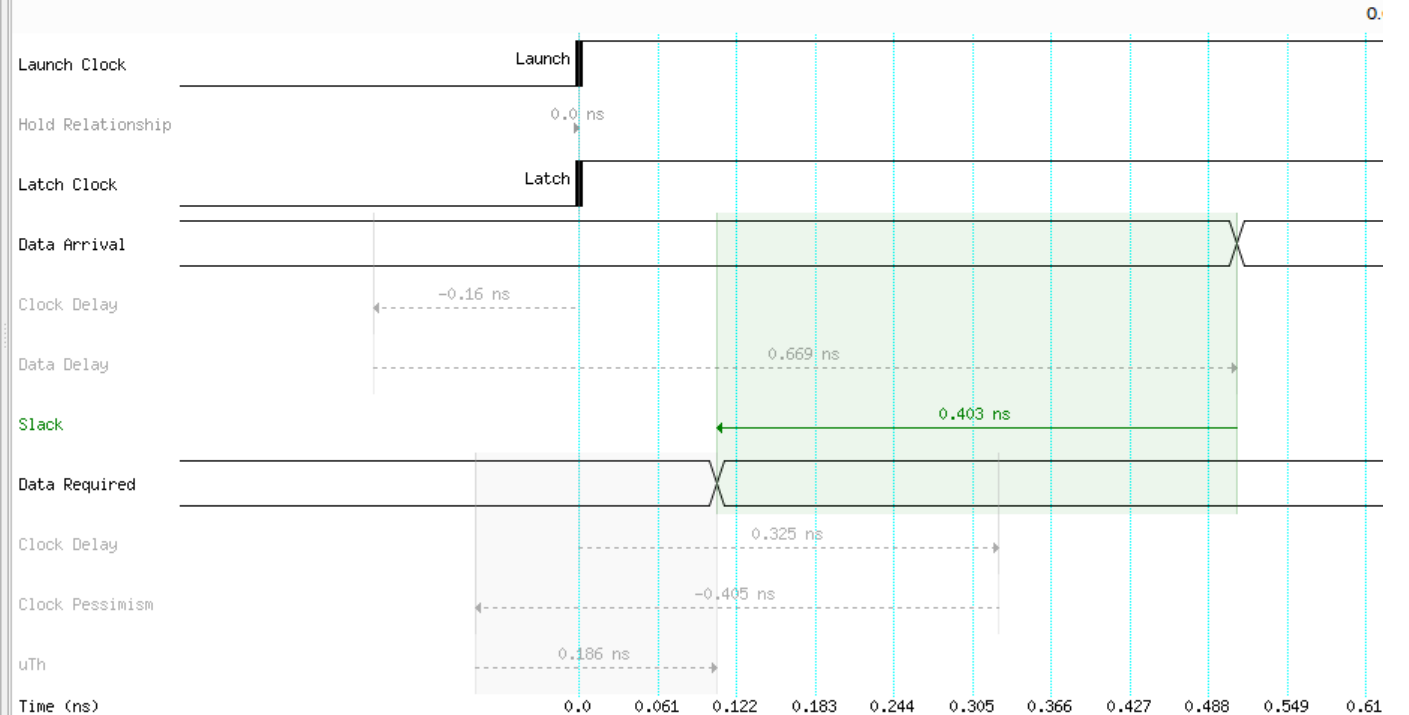
Path Summary

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Waveform

Extra Fitter Information



Counter_pll – fmax summary

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Slow 1200mV 85C Model Fmax Summary

<<Filter>>

	Fmax	Restricted Fmax	Clock Name	Note
1	268.02 MHz	268.02 MHz	m1 altpll_component auto_generated pll1 clk[0]	

This panel reports FMAX for every clock in the design, regardless of the user-specified clock periods. FMAX is only computed for paths where the source and destination registers or ports are driven by the same clock. Paths of different clocks, including generated clocks, are ignored. For paths between a clock and its inversion, FMAX is computed as if the rising and falling edges are scaled along with FMAX, such that the duty cycle (in terms of a percentage) is maintained. Altera recommends that you always use clock constraints a

Counter_pll - Power Summary

The screenshot displays the Power Analyzer Tool interface. The top bar shows two tabs: 'Compilation Report - Test' and 'Power Analyzer Tool'. The left sidebar, titled 'Table of Contents', lists various analysis categories. The 'Power Analyzer' category is expanded, and the 'Summary' item is selected and highlighted with a blue box. The main area on the right, titled 'Power Analyzer Summary', contains a search filter and a table of power analysis results.

Power Analyzer Summary	
Power Analyzer Status	Successful - Thu Apr 30 18:00:27 2026
Quartus Prime Version	21.1.0 Build 842 10/21/2021 SJ Lite Edition
Revision Name	Test
Top-level Entity Name	top
Family	Cyclone IV E
Device	EP4CE115F29C7
Power Models	Final
Total Thermal Power Dissipation	149.27 mW
Core Dynamic Thermal Power Dissipation	11.58 mW
Core Static Thermal Power Dissipation	101.67 mW
I/O Thermal Power Dissipation	36.02 mW
Power Estimation Confidence	Low: user provided insufficient toggle rate data